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MIL-B-80247C
20 February 1992
SUPERSEDING
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29 May 1987

MILITARY SPECIFICATION

BENDING MACHINES, SHEET AND PLATE, CYLINDRICAL, RESILIENT ROLLER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers forming machines constructed with resilient and steel forming rolls for bending metal workpieces into cylindrical shapes.

1.2 Classification. The bending machine shall be of the following types and sizes listed in tables I, II, and III. The types and sizes to be furnished shall be as specified (see 6.2).

Type I - Manually operated

Type II - Pneumatically operated

Type III - Hydraulically operated

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, TN 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN GEAR MANUFACTURERS' ASSOCIATION

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, Suite 201, 1500 King Street, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.12 - Roll - Forming and Roll - Bending Machines - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

NATIONAL FLUID POWER ASSOCIATION, INC. (NFPA)

NFPA/JIC T2.24.1 - Hydraulic fluid power - Systems standard for stationary industrial machinery.

NFPA/T2.25.1M - Pneumatic fluid power - Systems standard for industrial machinery.

(Application for copies should be addressed to the National Fluid Power Association Inc., 3333 N. Mayfair Road, Milwaukee, WI 53222-3291.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The machine shall be designed with not less than 2 forming rolls mounted in a horizontal plane, and shall be constructed so that performance requirements specified herein (see 3.6) are accomplished and maintained.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine roll positions shall be graduated in increments not larger than 1/32 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system shall be installed on the resilient roll axis of travel. The system shall consist of a position feedback device, an electronic display for the axis, and all related components, mounted hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of this axis.

3.2.2 Reclaimed materials . The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

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3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, electrical, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI B11.12. If a conflict arises between OSHA 29 CFR 1910 and ANSI B11.12, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the machine and its components, the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operators health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.7 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2), an automatic centralized lubrication system shall be provided.

3.2.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

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3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of axis drive trains shall conform or exceed all provisions of AGMA 390.03a, AGMA 2001.B88, and AGMA 2000-A88 the inch-pound system of measurement or ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a frame, rolls, roll position indicator, roll adjustment control, roll pressure lockup, roll drive system, upper roll support, and workpiece guide.

3.4.1 Frame. The frame shall be constructed with the required strength and rigidity to support and maintain alignment of all components of the machine while operating at its maximum capacity. Components supported by the frame shall be accessible for maintenance. Anchor bolt holes shall be provided to secure the machine to a mounting surface.

3.4.2 Rolls. The machine shall be equipped with one steel roll and one resilient roll. The steel roll shall have the hardness, surface finish, and toughness necessary to resist wear and prevent marring of the workpiece. The resilient roll shall be manufactured from a urethane elastomer having chemical and mechanical properties suitable for the intended use.

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3.4.2.1 Slip-on tubes. When specified (see 6.2), steel slip-on tubes for forming diameters larger than the steel roll shall be furnished. Slip-on tubes shall be of the length and diameter specified (see 6.2). Instructions for the development and use of slip-on tubes shall be furnished.

3.4.3 Roll position indicator. Unless otherwise specified (see 6.2), a mechanical roll position indicator, readable from the operator's position, shall be furnished to enable the operator to accurately locate and set the position of the resilient roll.

3.4.4 Roll adjustment control. The top roll shall be in a fixed position. The resilient roll shall be vertically adjustable based on the material, slip-on tube thickness (when required), and the required forming pressure. The resilient roll on the type I machines shall be manually adjustable. Unless otherwise specified (see 6.2), the resilient roll on the type II machines shall be pneumatically adjustable and the resilient roll on type III machines shall be hydraulically adjustable. Roll adjustment controls shall be located convenient to the operator at his normal work station.

3.4.4.1 Roll pressure lockup. The resilient roll shall have a locking device to apply and relieve the forming pressure between the rolls. The device shall relieve the roll pressure to facilitate workpiece removal. When the locking device is reengaged, the resilient roll shall return to the proper position to form a duplicate workpiece.

3.4.5 Roll drive system. The lower roll shall be driven either by an electric motor with reduction gear drive or a hydraulic drive system capable of providing smooth no-hesitant roll speeds conforming to the requirements of tables I, II, or III as applicable for the machine type required. The roll drive system shall have forward, reverse, and instant stop with controls located convenient to the operator at his normal work station.

3.4.6 Upper roll support. The upper roll shall be secured in the operating position by a stationary bearing support on one end and a retractable roll support on the free end. The roll support, when lowered, shall allow removal of the workpiece. The roll support for the type II machine shall be powered pneumatically and the type III machine hydraulically. The controls for the support shall prevent the rolls from making contact unless the support is in the operating position. A locking device shall be provided that secures the top roll and roll support in the operating position.

3.4.7 Workpiece guide. The machine shall be equipped with a workpiece guide of the manufacturer's standard commercial design normally furnished on the size machine supplied. The guide shall be used to square the side of the workpiece.

3.4.8 Pneumatic system. The pneumatic system for the type II machines shall be operated from an external air source and shall consist of not less than a regulator, filter, control valve, and the necessary cylinders, hoses, and piping. The pneumatic system shall conform to the requirements of NFPA/T2.25.1M.

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3.4.9 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of NFPA/JIC T2.24.1 and shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate.

TABLE I. Type I - manually operated machines, sizes and capacities.

Size	Working width (inches)	Diameter of steel roll (inches)	Rolling speed (feet per minute)	Material thickness gauge	Drive motor horsepower
18 x 3-1/2	18	3-1/2	16	16	1
24 x 3-1/2	24	3-1/2	16	18	1

NOTE: All requirements shall be not less than those shown.

TABLE II. Type II - pneumatically operated machines, sizes and capacities.

Size	Working width (inches)	Diameter of steel roll (inches)	Rolling speed (feet per minute)	Material thickness (gauge or inches)	Drive motor horsepower
18 x 3-1/2	18	3-1/2	20	14 ga.	1-1/2
18 x 4	18	4	20	11 ga.	1-1/2
18 x 5	18	5	20	3/16 in.	3
24 X 4	24	4	20	12 ga.	1-1/2
24 X 5	24	5	20	10 ga.	3
30 x 5	30	5	20	14 ga.	1-1/2

NOTE: All requirements shall be not less than those shown.

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TABLE III. Type III - hydraulically operated machines, sizes and capacities.

Size	Working width (inches)	Diameter of steel roll (inches)	Rolling speed (feet per minute)	Material thickness (gauge or inches)	Drive motor horsepower
18 x 6	18	6	18	3/16 in.	5
24 x 3	24	3-1/2	20	12 ga.	4
24 x 6	24	6	18	10 ga.	5
24 x 7	24	7	18	3/16 in.	5
30 x 6	30	6	18	11 ga.	3
30 x 7	30	7	18	10 ga.	5
30 x 8	30	8	18	3/16 in.	5
30 x 9-1/2	30	9-1/2	15	1/4 in.	7-1/2
36 x 6	36	6	18	12 ga.	3
36 x 8	36	8	18	10 ga.	5
36 x 9-1/2	36	9-1/2	15	1/4 in.	7-1/2
48 x 7	48	7	18	12 ga.	3
48 x 8	48	8	18	10 ga.	5
48 x 9-1/2	48	9-1/2	15	3/16 in.	7-1/2
48 x 12	48	12	12	3/16 in.	7-1/2
60 x 8	60	8	18	11 ga.	5
60 x 9-1/2	60	9-1/2	15	11 ga.	7-1/2
60 x 12	60	12	12	3/16 in.	10
72 x 8	72	8	18	16 ga.	5
72 x 9-1/2	72	9-1/2	15	11 ga.	7-1/2
72 x 12	72	12	12	9 ga.	10
89 x 9-1/2	89	9-1/2	15	14 ga.	7-1/2

NOTE: All requirements shall be not less than those shown.

3.4.10 Electrical requirements. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz (Hz), dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.10.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), each motor shall have a dripproof enclosure.

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3.4.10.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.11 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.4.12 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I, II or III for the machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The machine shall be capable of rolling metal workpieces of the thickness and width specified in table I, II, or III for the type and size machine specified, using mild steel of 50,000 p.s.i. tensile strength, into a cylinder with a minimum inside diameter of not more than one inch larger than the steel roll. The maximum diameter shall be limited only by work space above the machine. When tested in accordance with 4.7.3.1, the formed cylinder shall have no visible damage to the surface contacted by the resilient roll. The contour of the cylinder shall deviate not more than 0.5 percent excluding the leading and trailing edge flats. The edge flats on the formed workpiece shall be not greater than four times material thickness. The leading and trailing edges of the formed workpiece shall be parallel within 0.125 inch per foot of rolled cylinder center line.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified and fully described (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or machine.

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3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.10.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.11 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.

3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the test(s) in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

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4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. Each machine shall be operated at no load for not less than 1/2 hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.3 Performance tests. Each machine shall form a workpiece of the capacity width and thickness shown in tables I, II, or II for the type and size machine specified, into a cylinder having an inside diameter not more than 1 inch larger than the steel roll. The cylinder shall be formed in one pass. The contour, parallelism of edges, and end flats shall conform to the requirements of 3.6. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.3.1 Surface damage test. The surface damage test shall be performed using a polished, annealed aluminum sheet not less than the machine's effective working width with a thickness not less than 18 gauge. The workpiece shall be rolled to form a cylinder of the minimum diameter for the machine tested. There shall be no visible evidence of scratches or abrasions of the finished surface due to contact with the resilient roll.

4.7.3.2 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.7.3 shall be as specified and fully described (see 6.2).

4.7.4 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.1. Unless otherwise specified (see 6.2), the measurements shall be made at the operators work station and each corner of the machine. Noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response. The test shall be conducted with the machine operating at normal speeds.

4.7.5 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.7.6 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Machines covered by this specification are intended for use in metal rolling and bending operations that are generally required in sheet metal fabrication work.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Digital readout system, specify and fully describe if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Noise level, if different (see 3.2.5.1).

i. Environmental protection requirements exceeding EPA/OSHA requirements (see 3.2.6).

j. Automatic centralized lubrication system, if required (see 3.2.7).

k. Painting, if different (see 3.3.5.).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

m. Steel slip-on tubes for forming larger diameters than the steel roll, if required (see 3.4.2.1).

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n. Length and diameter of slip-on tubes, if required (see 3.4.2.1).

o. Roll position indicator, if required (see 3.4.3).

p. Resilient roll adjustment, if different (see 3.4.4).

q. Voltage requirements, if different (see 3.4.10).

r. Initial voltage, as specified (see 3.4.10).

s. Motors, if different (see 3.4.10.1).

t. Worklight, if required (see 3.4.11).

u. Hour meter, if required (see 3.4.12).

v. Size and capacity, if different (see 3.5).

w. Specify physical size limitation, if required (see 3.5.1).

x. Additional performance features, if required (see 3.6.1).

y. Specify and fully describe optional equipment, if required (see 3.8).

z. Specify applicable repair parts, if required (see 3.9).

aa. Lubrication plate, if required (see 3.10.1).

bb. Lubrication plate information, if different (see 3.10.1).

cc. Nameplate, if different (see 3.10.2).

dd. Hydraulic fluid requirements plate, if required (see 3.10.3).

ee. Hydraulic fluid plate information, if different (see 3.10.3).

ff. Inspection conditions, if different (see 4.3).

gg. First article inspection, if different (see 4.4).

hh. Quality conformance inspection, if different (4.5).

ii. Additional performance tests,if required (see 4.7.3.2).

jj. Noise level acceptance test, if different (see 4.7.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of

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the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

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6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-B-80247B	MIL-B-80247C
Type II	Type II Size 18x4 (new size)
Type III	Type III Size 30x8 (new size) Size 48x8 (new size) Size 60x8 (new size) Size 72x8 (new size)

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

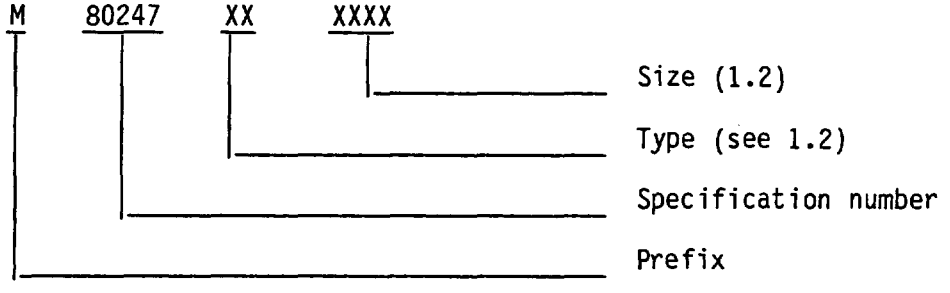
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PINs to be used for machines acquired to this specification are created as follows:



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6.14 Subject term Key word listing.

Bending
Cylindrical
Forming
Machine
Metal, sheet, bending, drawing, working
Resilient
Roll, forming

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Navy - MC

Preparing Activity:

DLA - IP

Project (3441-0150)

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